

Payment Processing Data Engineering Pipeline

An End-to-End AWS Data Lake for Fraud, Dispute, and Compliance Analytics

INSS 661 — Data Engineering | Morgan State University | December 2025

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Overview

This project designed and built a complete cloud data engineering pipeline for a payment-processing scenario, taking raw, multi-source financial data all the way through to governed, analytics-ready datasets and interactive dashboards. The pipeline runs on Amazon Web Services using a Medallion (Landing → Clean → Curated) data-lake architecture, and it supports three real needs in digital payments: fraud detection, dispute and refund resolution, and regulatory compliance reporting.

It covers the full data-engineering lifecycle — ingestion, storage, transformation, cataloging, dimensional modeling, serverless SQL querying, and visualization — showing how well-built data infrastructure turns fragmented raw data into decision-ready intelligence for downstream users such as machine-learning engineers, data and compliance analysts, and customer-support teams.

Data Modeling

The source data was first modeled as a normalized relational schema (OLTP, 3NF), then restructured into a star schema (OLAP) optimized for analytics and BI querying.

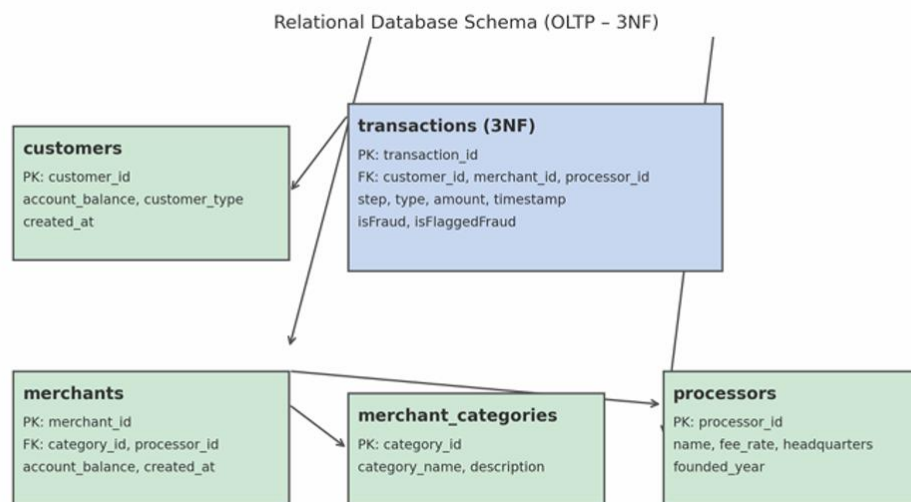


Figure 1 — Source relational schema (OLTP, third normal form).

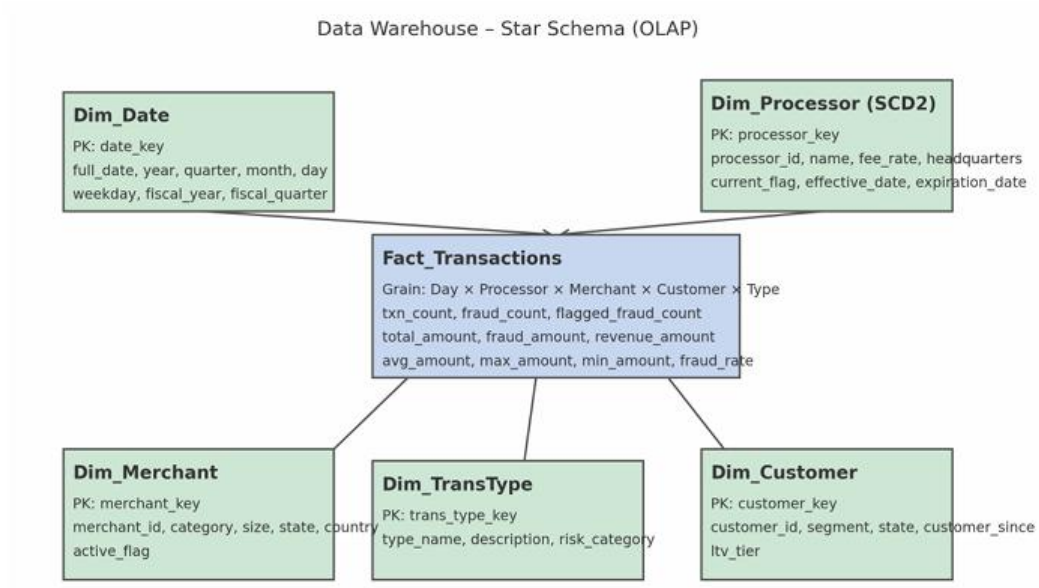


Figure 2 — Analytics data model (star schema, OLAP).

System Architecture

The pipeline uses a three-zone Medallion architecture across Amazon S3, with AWS Glue for transformation and cataloging, AWS Lambda for automation, Amazon Athena for querying, and Tableau for visualization.

Data Lake Zones (Amazon S3)

- **Landing Zone (Raw):** source files stored in native XLS/CSV format, full lineage preserved, no schema enforcement.
- **Clean Zone (Refined):** validated, standardized data converted to Parquet (Snappy compression), cataloged as a Glue database.
- **Curated Zone (Analytics-ready):** modeled, joined datasets optimized for downstream querying and BI.

Data Storage & Cataloging

Raw files were ingested into S3 and converted to Parquet through AWS Glue. Lambda triggers automatically updated the Glue Data Catalog as new data arrived. The Clean Zone database held six governed tables: transaction_data, crm_support_data, financial_data, fed_payments, mastercard_benchmarks, and visa_fraud_benchmarks.

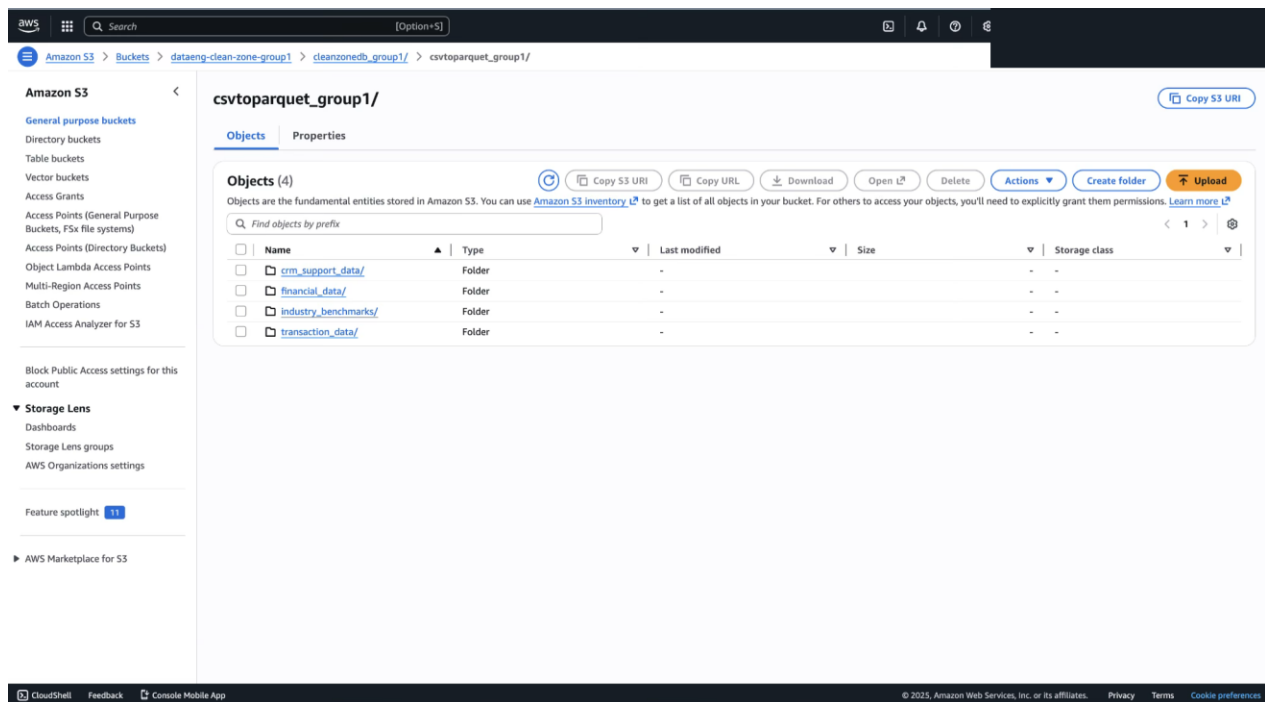


Figure 3 — Clean Zone database and tables in the AWS Glue Data Catalog.

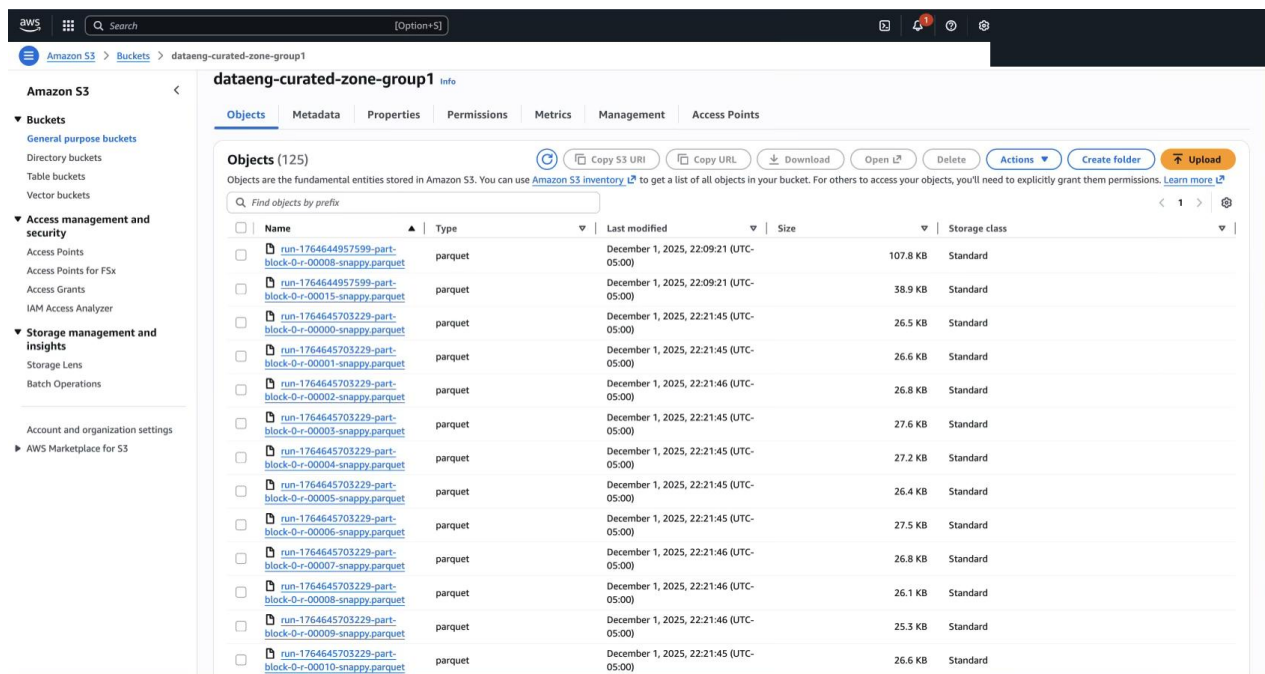


Figure 4 — Cataloged tables and schemas in AWS Glue.

Transformation & Data Modeling

Transformations were carried out in AWS Glue Studio using both visual transforms and SQL — type enforcement, deduplication, null handling, schema standardization, value/code standardization ("Yes"/"No" → 1/0; billions/trillions → millions), period splitting, and multi-table joins into a consolidated final_joined_dataset.

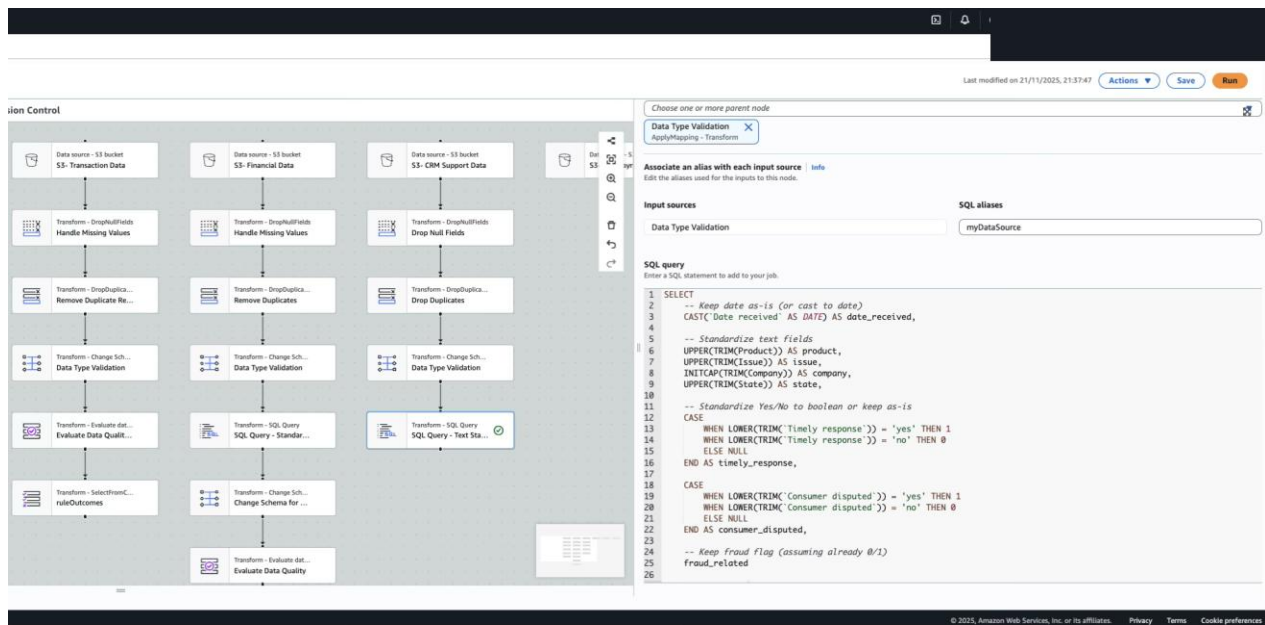


Figure 5 — AWS Glue Studio visual ETL job.

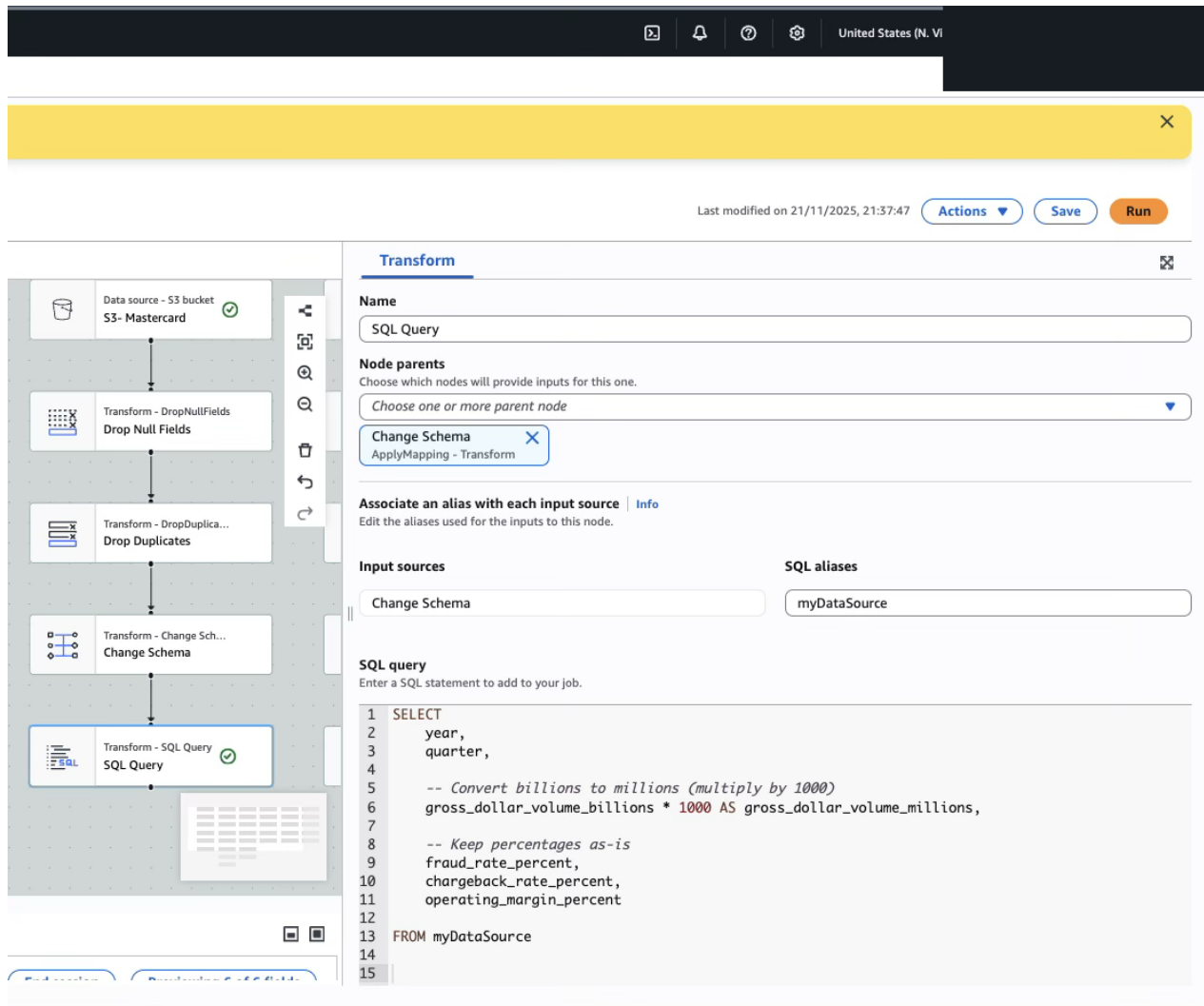


Figure 6 — SQL transformation in Glue Studio (value/text standardization).

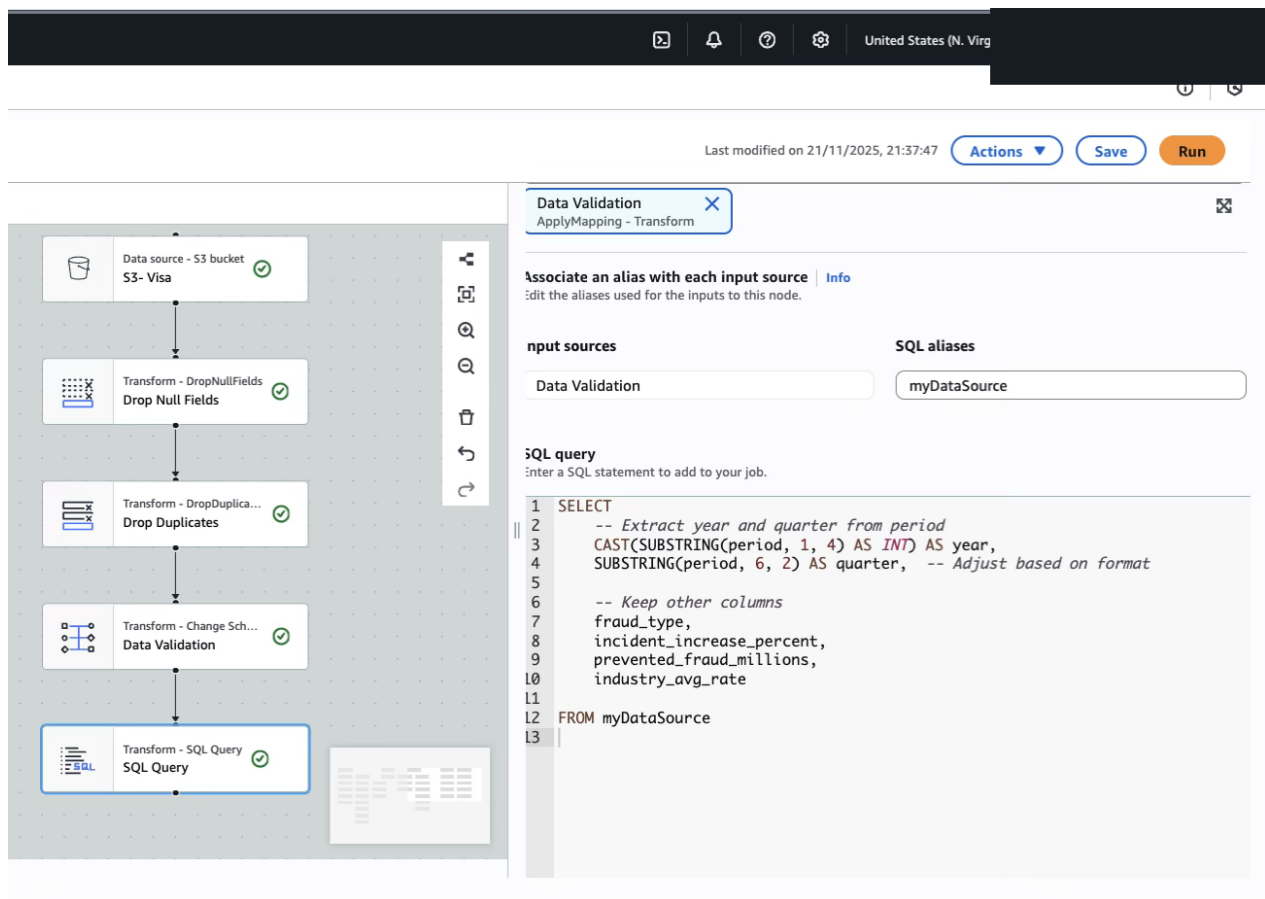


Figure 7 — Glue Studio join workflow producing the final joined dataset.

Querying (Amazon Athena)

Analytical querying was performed in Amazon Athena using serverless SQL directly on the curated Parquet tables, integrated with the Glue Data Catalog.

The Amazon Athena query editor shows the following details:

- Query 3:** `SELECT CASE WHEN timely_response = 1 THEN 'Timely Response' WHEN timely_response = 0 THEN 'Delayed Response' ELSE 'Unknown' END AS response_category, COUNT(*) AS ticket_count, SUM(CASE WHEN is_fraud_related = 1 THEN 1 ELSE 0 END) AS fraud_related_tickets, SUM(CASE WHEN consumer_disputed = 1 THEN 1 ELSE 0 END) AS disputed_tickets, ROUND(AVG(CASE WHEN consumer_disputed = 1 THEN 1.0 ELSE 0.0 END) * 100, 2) AS dispute_rate_percent FROM crm_data_cleaned GROUP BY timely_response;`
- Query status:** Completed
- Results (3):**

#	response_category	ticket_count	fraud_related_tickets	disputed_tickets	dispute_rate_percent
1	Timely Response	9207	3717	1075	11.68
2	Delayed Response	701	258	90	12.84
3	Unknown	155669	0	0	0.0

Figure 8 — Amazon Athena SQL query and results over the curated tables.

The screenshot displays the Amazon Athena console interface. On the left, a workflow diagram shows a sequence of nodes: 'Data source - S3 bucket S3- Fed_payments', 'Transform - DropNullFields Drop Null Fields', 'Transform - DropDuplica... Drop Duplicates', 'Transform - Change Sch... Change Schema', and 'Transform - SQL Query SQL Query - Standar...'. The right panel shows the configuration for the 'SQL Query' node. It includes a 'Choose one or more parent node' dropdown, a 'Change Schema' button, and a section for 'Associate an alias with each input source'. Below this, the 'Input sources' section shows 'Change Schema' and the 'SQL aliases' section shows 'myDataSource'. The 'SQL query' section contains the following SQL statement:

```
1 SELECT
2   year,
3   payment_method,
4
5   -- Convert billions + millions
6   total_transactions_billions * 1000 AS total_transactions_millions,
7
8   -- Convert trillions + millions
9   total_value_trillions * 1000000 AS total_value_millions,
10
11   fraud_rate_percent,
12
13   -- Already in millions, just passthrough
14   fraud_loss_millions
15
16 FROM myDataSource;
17
```

The bottom of the console shows the copyright notice: '© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

Figure 9 — Athena query for a payment-processing business question.

Analysis & Key Findings

Problem 1 — Fraud Detection

- DEBIT transactions had the highest fraud rate at **0.75%**; TRANSFER carried the greatest exposure at **\$15,295.91** in losses (0.59%); PAYMENT was safest at 0.22%.

Fraud Rate and Financial Impact by Transaction Type

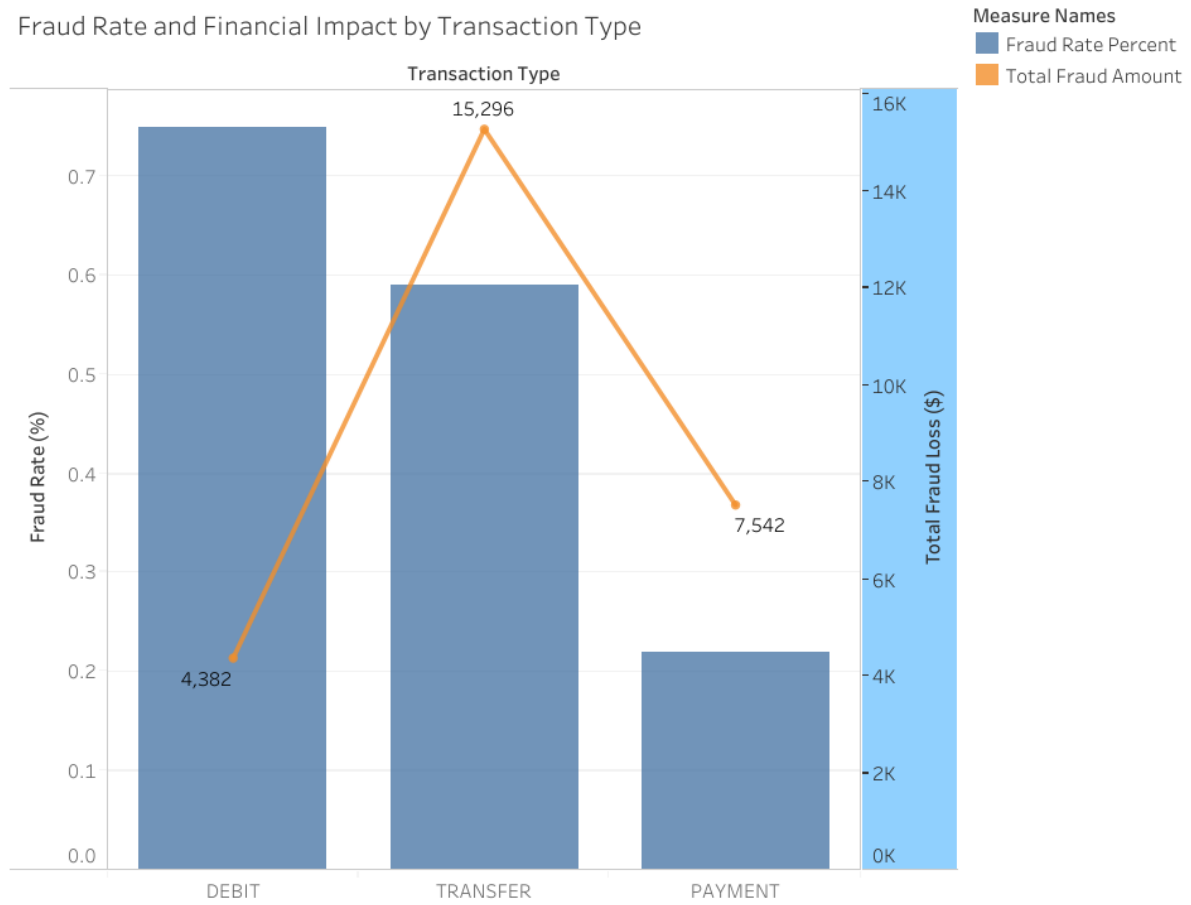


Figure 10 — Tableau: fraud rate and financial impact by transaction type.

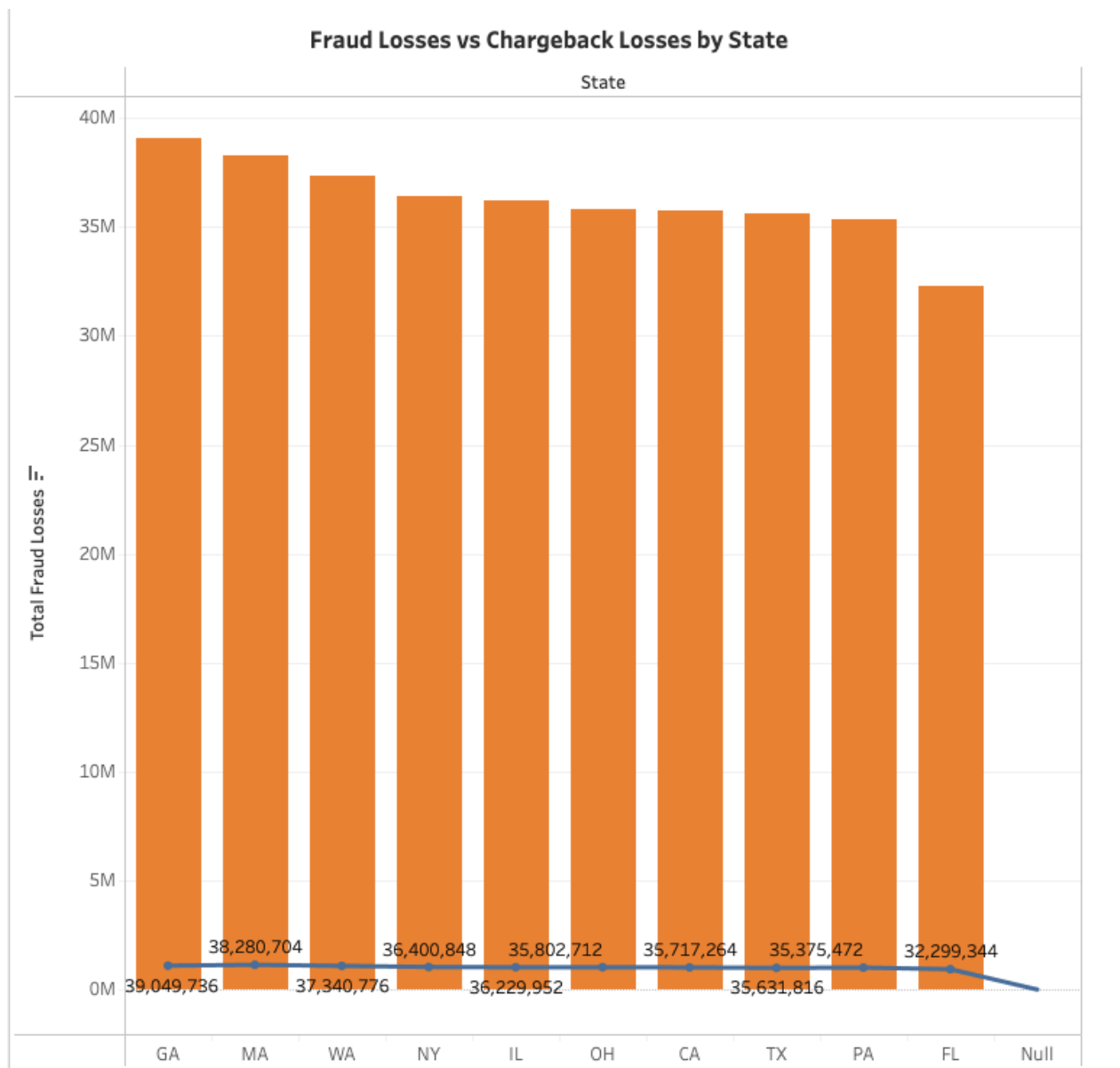


Figure 11 — Tableau: fraud vs. chargeback losses by state.

Problem 2 — Dispute & Refund Resolution

- Billing disputes were highest-risk for Money Transfer products at **13.64%**; Credit Card products showed elevated duplicate-charge (12.70%) and billing (12.64%) disputes.

Customer Dispute Rates by Product and Issue Type

Product	Issue						Dispute Rate %
	Billing	Duplicate	Fraud/Scam	No Refund	No Value Received	Unauthorized	
Credit Card/Prepaid Card	12.640	12.700	11.860	12.040	11.780	10.700	10.520 13.640
	134	87	97	120	65	149	
Money Transfer/Virtual Currency	13.640	10.520	11.520	12.030	10.630	10.870	
	118	57	75	95	47	121	

Figure 12 — Tableau: customer dispute rates by product and issue type (heat map).

Problem 3 — Compliance & Regulatory Reporting

- Visa PERC trends showed a worsening 2024 landscape: **Enumeration +25%**, Account Takeover +20%, Card-Not-Present +14%, with **\$760M prevented in Q4**.

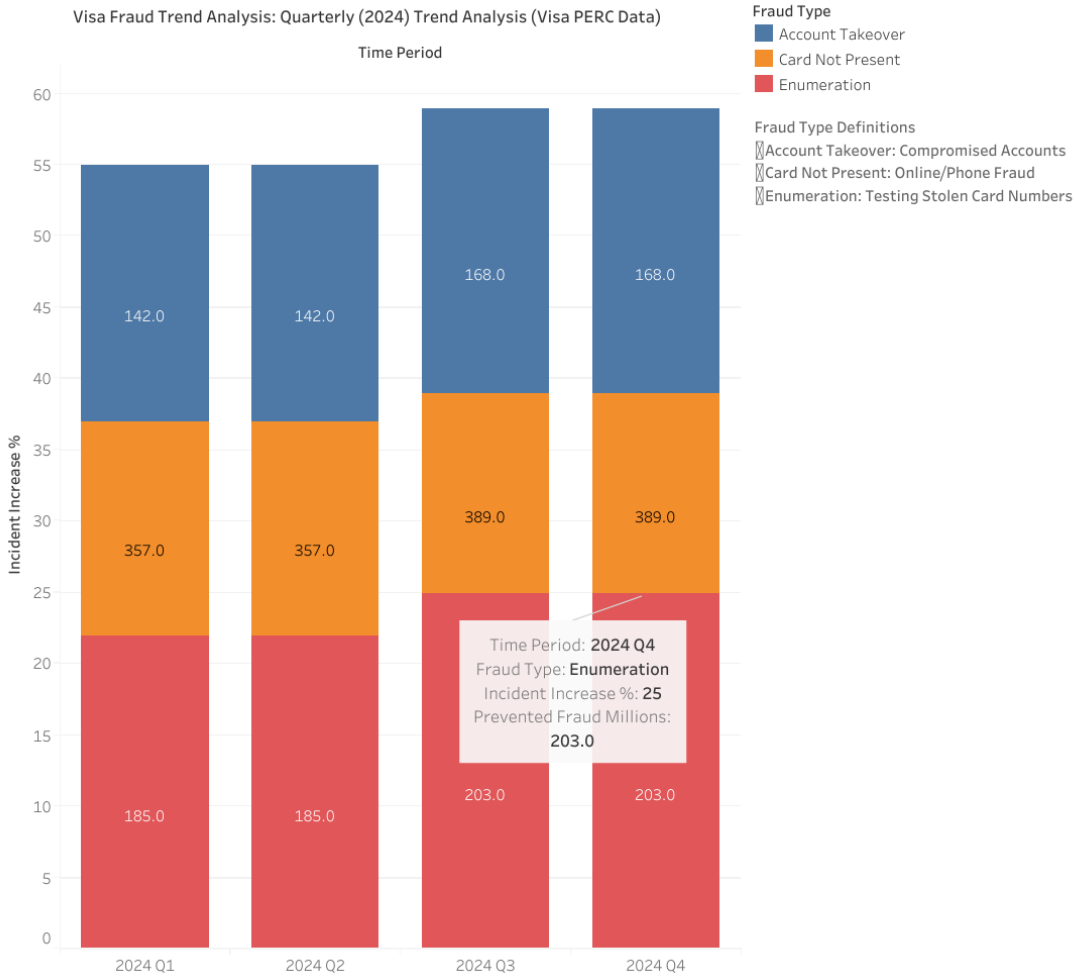


Figure 13 — Tableau: quarterly fraud-trend analysis (Visa PERC data).

Consolidated Dashboard

All three problem areas were brought together into a single interactive Tableau dashboard for a unified view across fraud, disputes, and compliance.

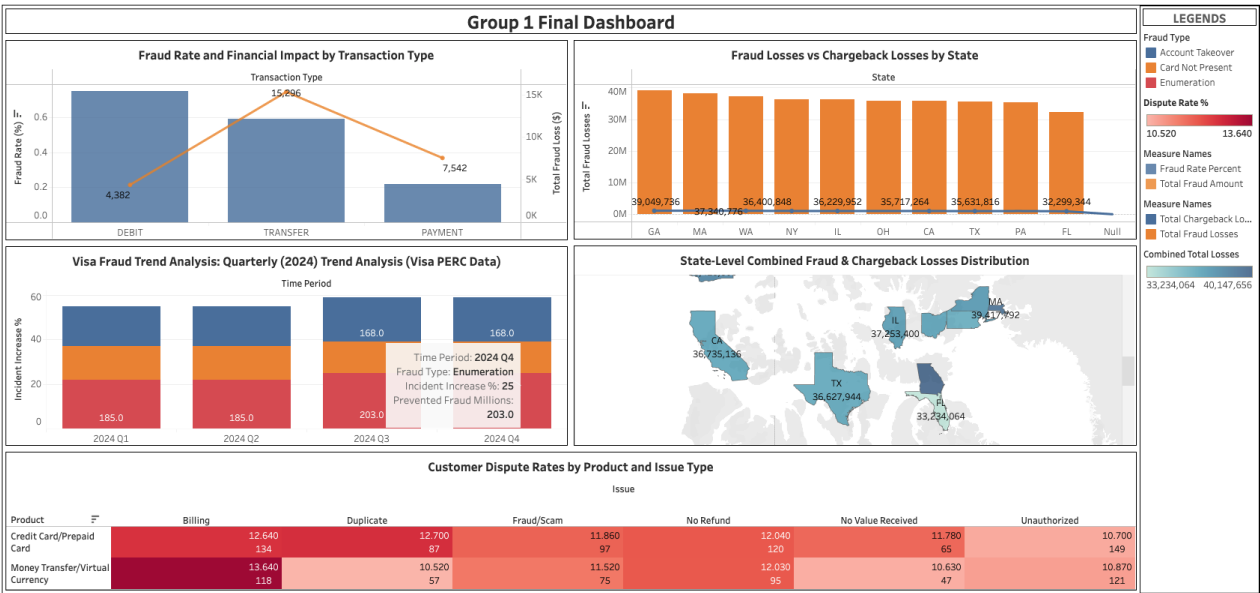


Figure 14 — Final consolidated Tableau dashboard.

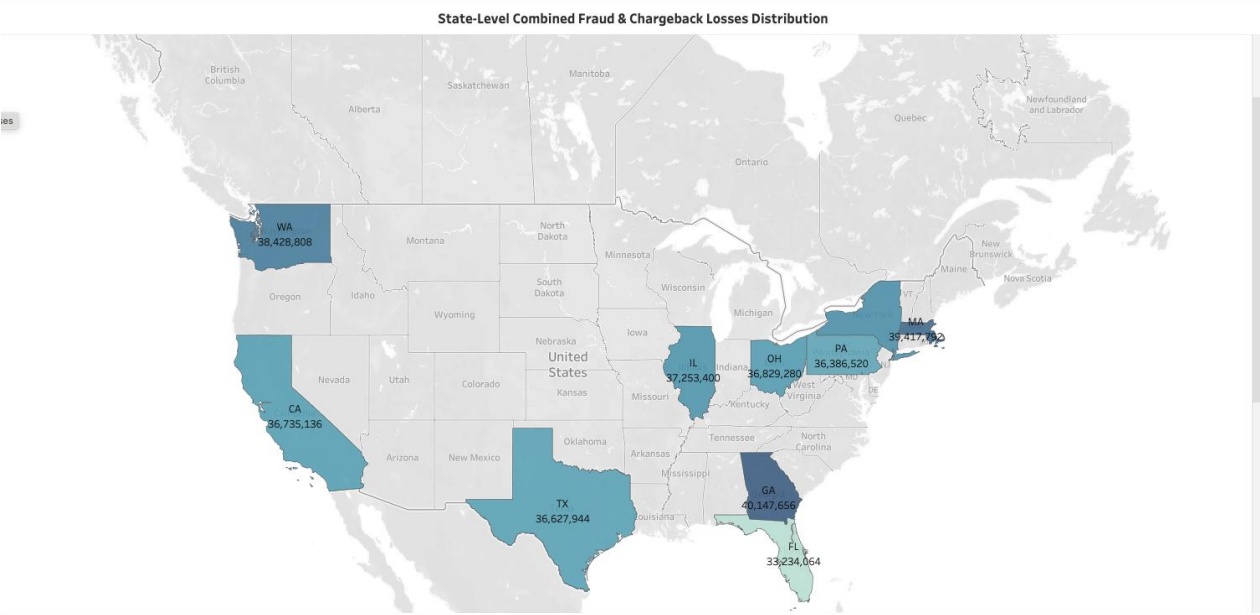


Figure 15 — Geographic view: fraud exposure by U.S. state.

Technologies & Skills Demonstrated

A r e a	T o o l s / C o n c e p t s
C l o u d d a t a e n g i n e e r i n g	A W S S 3 , A W S G l u e (S e t u d i o + D a t a C a t a l o g) , A W S L a m b d a ,

A r e a	T o o l s / C o n c e p t s
	A m a z o n A t h e n a
A r c h i t e c t u r e	M e d h a l l e c t o n d r a t a l a k e (L a n d i n g / C l e a n /

A r e a	T o o l s / C o n c e p t s
	C u r a t e d) ; s e r v e r l e s s , e v e n t - d r i v e n a u t o m a t i o n
D a	D C L

A r e a	T o o l s / C o n c e p t s
t a n o d e l i n g	T P 3 N F a n d C L A P s t a r s c h e m a ; d e n o r m a l i z a t i o n a n d m u l

A r e a	T o o l s / C o n c e p t s
	t i - t a b l e j o i n s
E X T R A & q u a l i t y	E X T R A - P a r t i c u l a r s c h e m a t i c o r r e c t i o n

A r e a	T o o l s / C o n c e p t s
	e n t , d e d u p l i c a t i o n , n u l l h a n d l i n g , s t a n d a r

A r e a	T o o l s / C o n c e p t s
	d i z a t i o n
Q u e r y i n g	S Q L o n A m a z o n A t h e n a o v e r c u r a t e d P a r q u e

A r e a	T o o l s / C o n c e p t s
	t t a b l e s
G o v e r n a n c e	D e a t h l i n e a n c e g e , c a t a l o g i n g , a n d a u d i t a b l e p i

A r e a	T o o l s / C o n c e p t s
	p e r i n e s (A M L / P C I - D S S c o n t e x t)
V i s u a l i z a t i o n	T a b u l e a u v a i u s o n l s a n d

A r e a	T o o l s / C o n c e p t s
	i n t e r a c t i v e d a s h b o a r d e s i g n

Team & Contributions

This was a collaborative Group 1 project; all three members contributed across the pipeline. **Ayomide Ajibola** contributed to the data pipeline and analysis and focused on the visualization work — developing the Tableau visuals and dashboard that translated the curated query outputs into decision-ready insights across the fraud, dispute, and compliance problem areas.